$\begin{bmatrix} 2 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 2 & 0 & 0 \\ 4 & 1 & 0 \\ 8 & -5 & 7 \end{bmatrix}$, $\begin{bmatrix} -2 & 0 \\ 9 & -3 \end{bmatrix}$, and $\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ are all lower triangular matrices.

Definition 7.12

A square matrix which is either upper triangular or lower triangular is called a **triangular matrix**.

Observe that a square matrix that is both upper and lower triangular simultaneously will turn out to be a diagonal matrix.

7.2.2 Equality of Matrices

Definition 7.13

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are **equal** (written as $A = B$) if and only if

(i) both A and B are of the same order

(ii) the corresponding entries of A and B are equal. That is, $a_{ij} = b_{ij}$ for all i and j.

For instance, if

$$\begin{bmatrix} x & y \\ u & v \end{bmatrix} = \begin{bmatrix} 2.5 & -1 \\ \frac{1}{\sqrt{2}} & \frac{3}{5} \end{bmatrix}, \text{ then we must have } x = 2.5, y = -1, u = \frac{1}{\sqrt{2}} \text{ and } v = \frac{3}{5}.$$

Definition 7.14

Two matrices A and B are called **unequal** if either of condition (i) or (ii) of Definition 7.13 does not hold.

For instance, $\begin{bmatrix} 4 & -3 \\ 0 & 8 \end{bmatrix} \neq \begin{bmatrix} 8 & -5 \\ 0 & 4 \end{bmatrix}$ as the corresponding entries are not equal. Also $\begin{bmatrix} 4 & -3 \\ 0 & 8 \end{bmatrix} \neq \begin{bmatrix} 5 & -8 \\ 3 & 4 \\ 6 & 7 \end{bmatrix}$ as the orders are not the same.

Example 7.3

Find x, y, a, and b if $\begin{bmatrix} 3x+4y & 6 & x-2y \\ a+b & 2a-b & -3 \end{bmatrix} = \begin{bmatrix} 2 & 6 & 4 \\ 5 & -5 & -3 \end{bmatrix}$.

Solution

As the orders of the two matrices are same, they are equal if and only if the corresponding entries are equal. Thus, by comparing the corresponding elements, we get

$3x+4y=2$, $x-2y=4$, $a+b=5$, and $2a-b=-5$.

Solving these equations, we get $x=2$, $y=-1$, $a=0$, and $b=5$.

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